

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T54 · Histograms

With Answers

Questions with full worked solutions

■ 41 pages

FAMILY MAP

Histograms

F01	Interpret histogram areas	11 Q 37 marks
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F02	Construct and complete histograms	5 Q 13 marks
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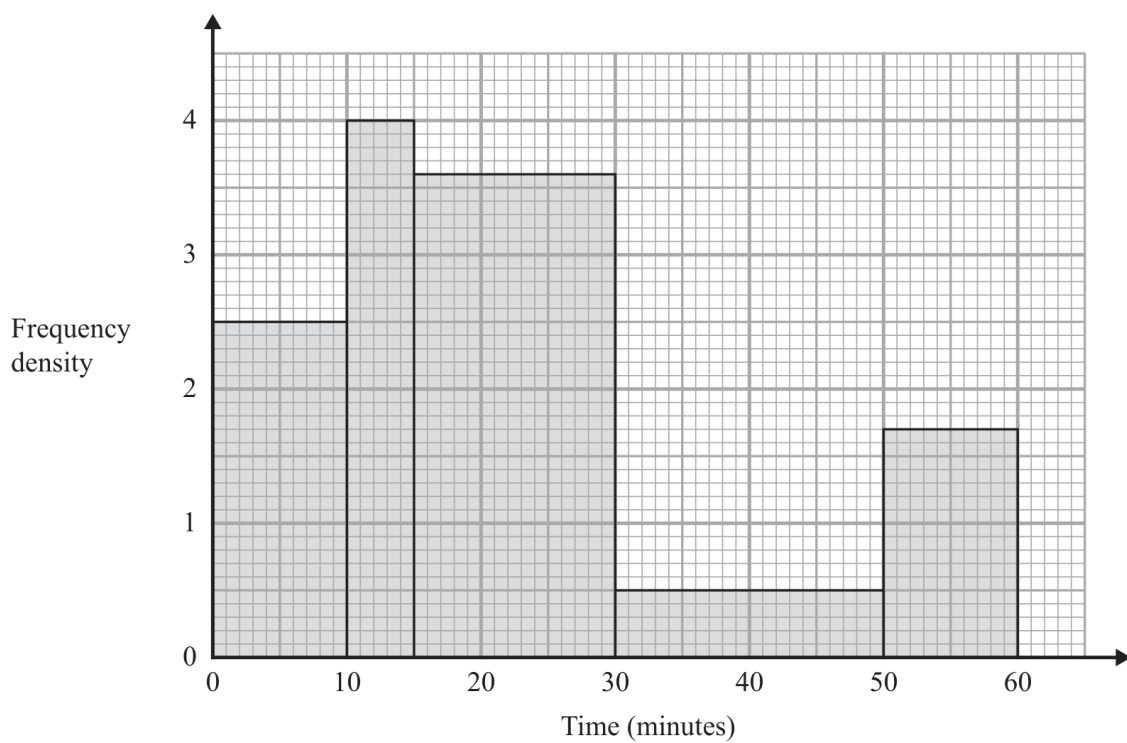
F03	Use the frequency-density model	1 Q 3 marks
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Working Unit 12

4MA1-1H Q17

ORIGINAL QUESTION**3 marks**

- 17 The histogram shows information about the times, in minutes, that some people were in a shop.



Work out an estimate for the proportion of these people who were in the shop for more than 40 minutes.

(Total for Question 17 is 3 marks)

Worked Solution 12

Estimate a proportion from a histogram

UNIT 1
3 marks**Find frequencies from areas**

$$10(2.5) = 25, 5(4) = 20, 15(3.6) = 54, 20(0.5) = 10, 10(1.7) = 17.$$

Total frequency

$$25 + 20 + 54 + 10 + 17 = 126.$$

More than 40 minutes

$$10(0.5) + 10(1.7) = 5 + 17 = 22, \quad \frac{22}{126} = \frac{11}{63} \approx 0.175.$$

Final answer

$$\frac{11}{63} \approx 0.175$$

Dependency check Only the 40–50 half of the 30–50 class is included.

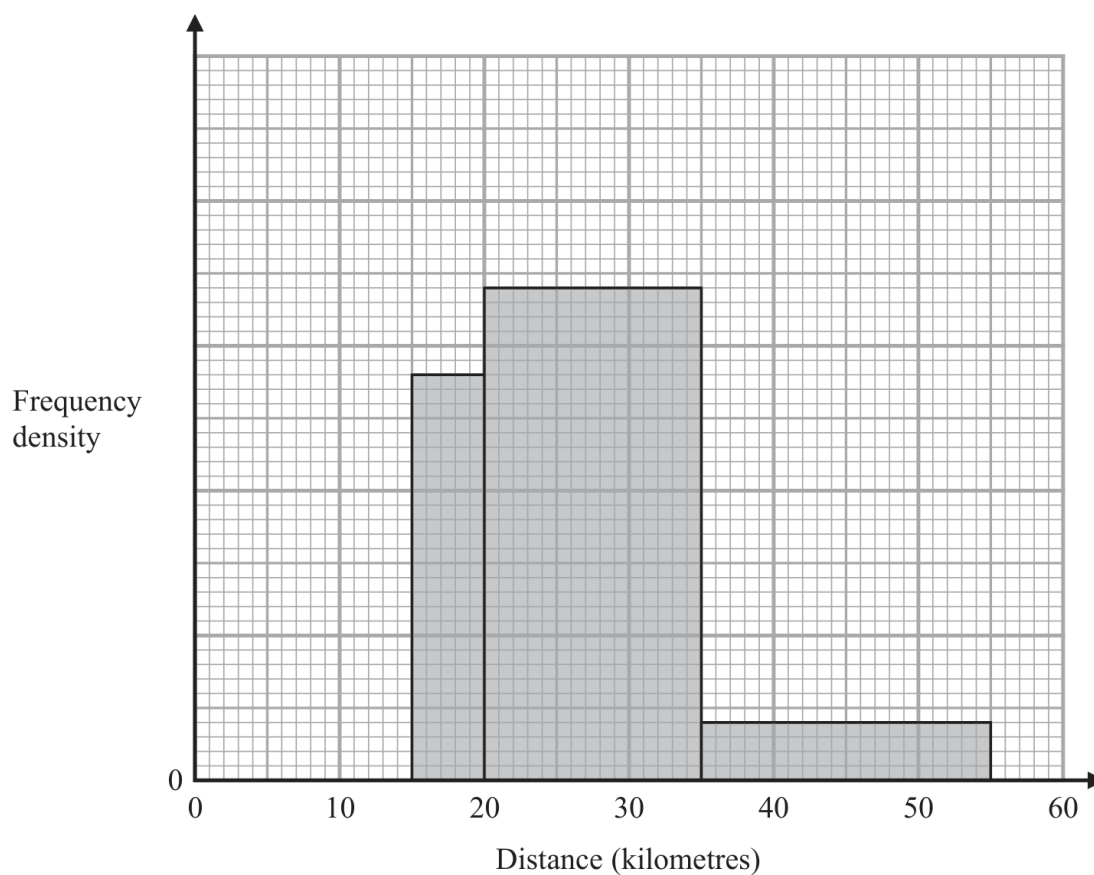
Question 15

4MA1-1H Q22

ORIGINAL QUESTION

3 marks

- 22 The incomplete histogram shows some information about the distances, in kilometres, that 100 adults ran last week.



All of the adults ran at least 5 kilometres.

None of the adults ran more than 55 kilometres.

14 adults ran between 15 kilometres and 20 kilometres.

Complete the histogram.

Worked Solution 15

Complete a histogram

MODULAR UNIT 1

3 marks

Scale

$$15 \leq d < 20 : 14/5 = 2.8 \text{ frequency density.}$$

Missing class

$$\text{Known frequency} = 14 + 51 + 8 = 73, \quad 100 - 73 = 27.$$

Bar

$$5 \leq d < 15 \text{ has density } 27/10 = 2.7.$$

Final answer

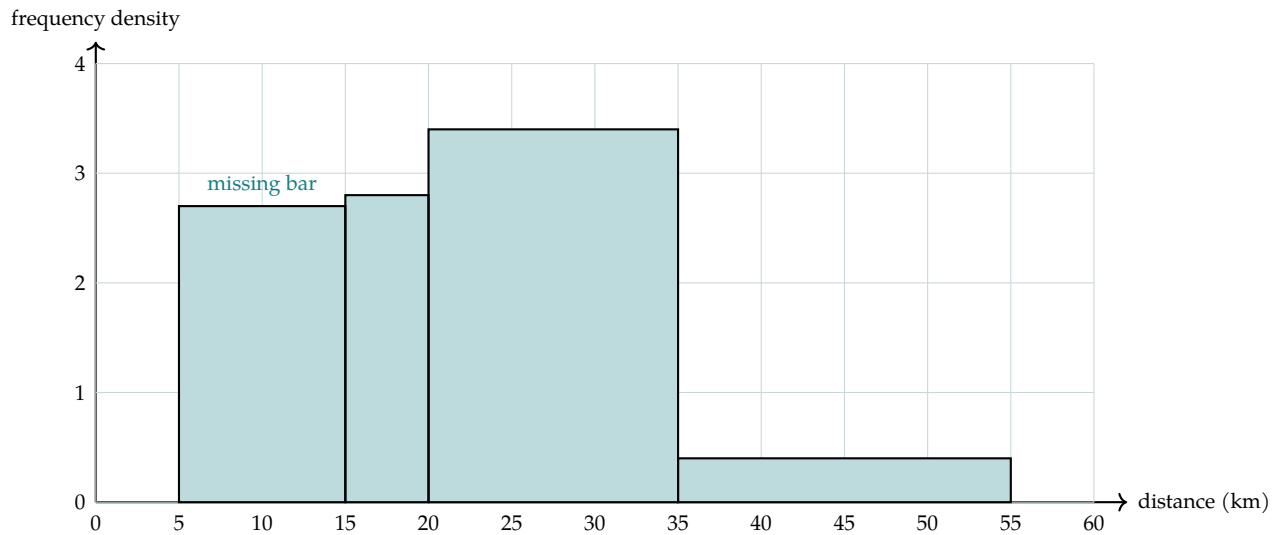
Draw the $5 \leq d < 15$ bar at density 2.7

Completed diagram 15

Complete a histogram

MODULAR UNIT 1

3 marks

Completed solution diagram: Completed histogram

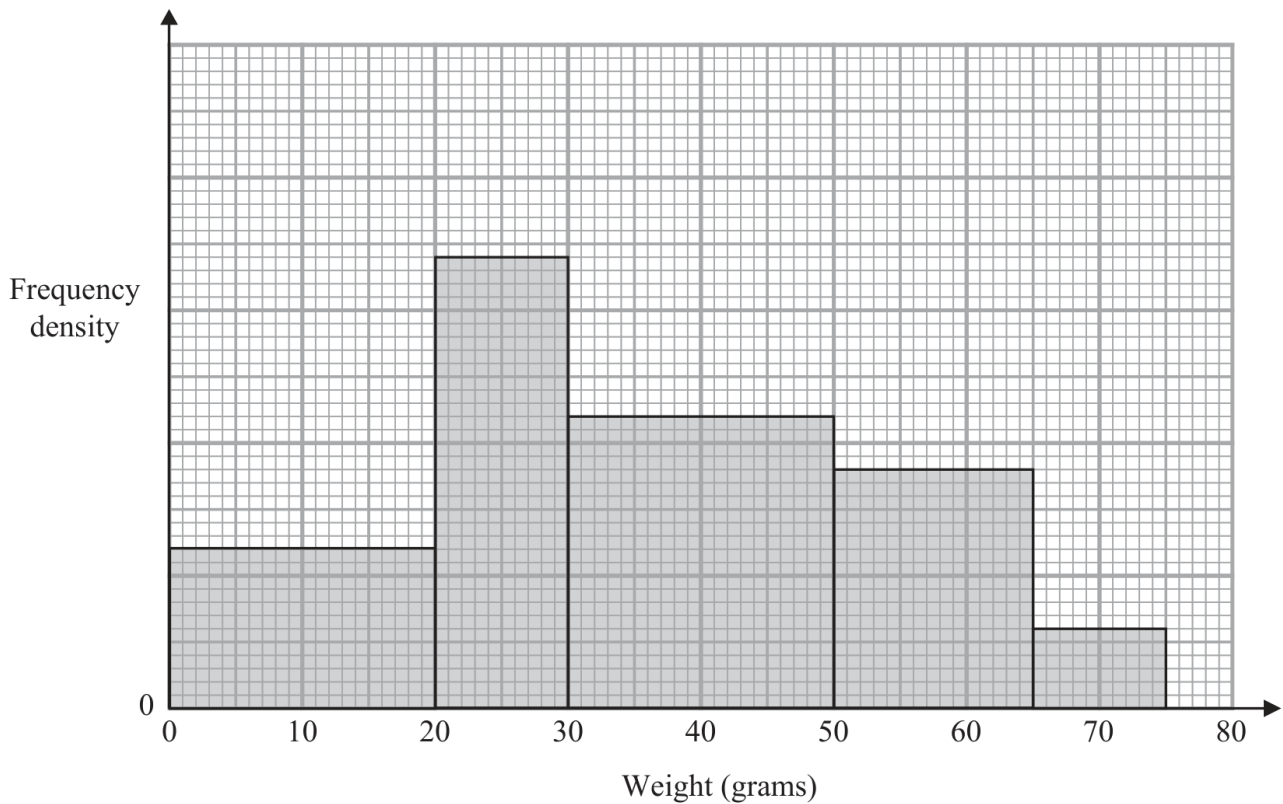
Question 13

4MA1-1H Q17 demand

ORIGINAL QUESTION UNIT 1

3 marks

17 The histogram gives information about the weights, in grams, of some oranges in a box.



24 of these oranges weigh less than 20 grams.

Medium oranges weigh between 35 grams and 55 grams.

Work out an estimate for the number of medium oranges in the box.

.....

Worked solution 13

Histogram area estimate

MODULAR UNIT 1

3 marks

Demand – Histogram area estimate

Calibrate the density scale

 $24/20 = 1.2$; the first bar therefore sets the small – square scale.

Add the two areas

$$35-50 : 15(2.2) = 33; \quad 50-55 : 5(1.8) = 9.$$

Total

$$33 + 9 = 42.$$

Final answer

42 oranges

Question 10

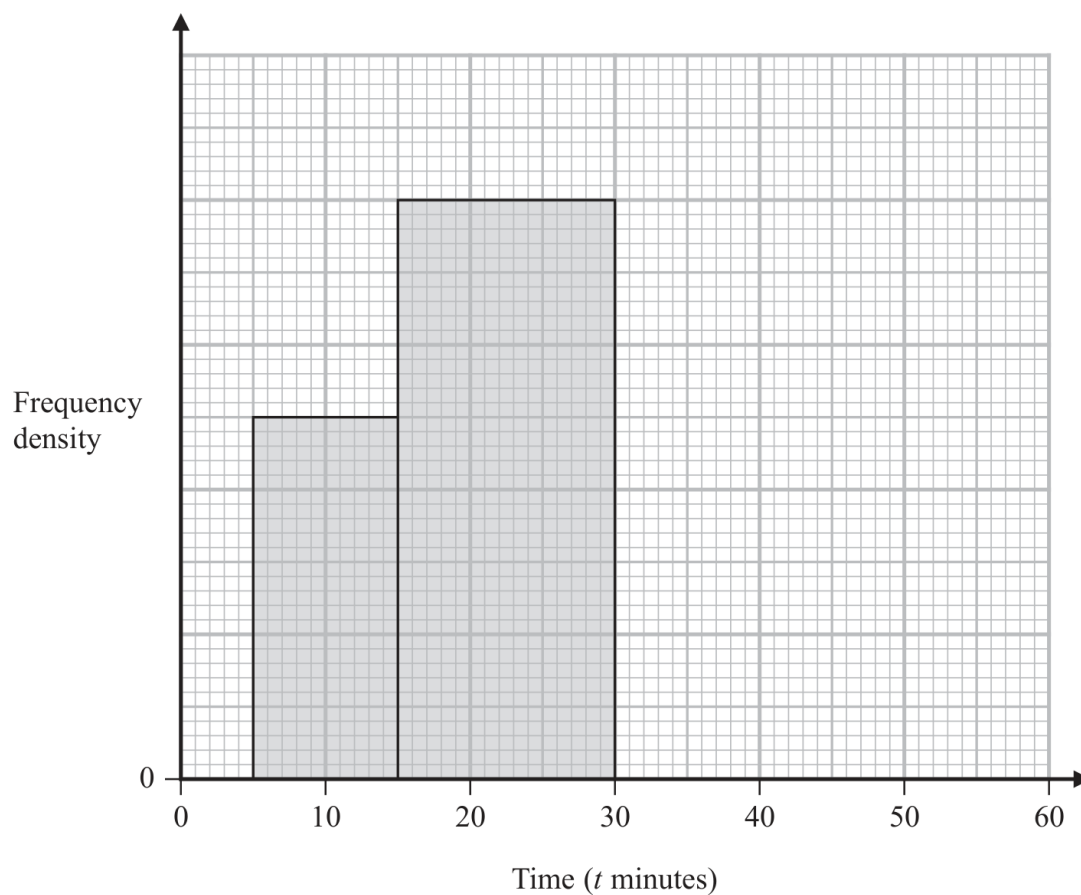
4MA1-1H Q18

ORIGINAL QUESTION

4 marks

- 18 The incomplete table and incomplete histogram give information about the times, in minutes, that 140 people waited at a station for a train.

Time (t minutes)	Frequency
$0 < t \leq 5$	23
$5 < t \leq 15$	
$15 < t \leq 30$	
$30 < t \leq 40$	18
$40 < t \leq 60$	14



Complete the table and the histogram.

Worked Solution 10

Complete a grouped table and histogram

MODULAR UNIT 1

4 marks

Find the two missing frequencies

$$140 - (23 + 18 + 14) = 85, \quad 85 \text{ is shared in the ratio } 10(2.5) : 15(4) = 25 : 60.$$

Complete the table

23, 25, 60, 18, 14.

Use frequency density

$$\frac{23}{5} = 4.6, \quad \frac{25}{10} = 2.5, \quad \frac{60}{15} = 4, \quad \frac{18}{10} = 1.8, \quad \frac{14}{20} = 0.7.$$

Final answer

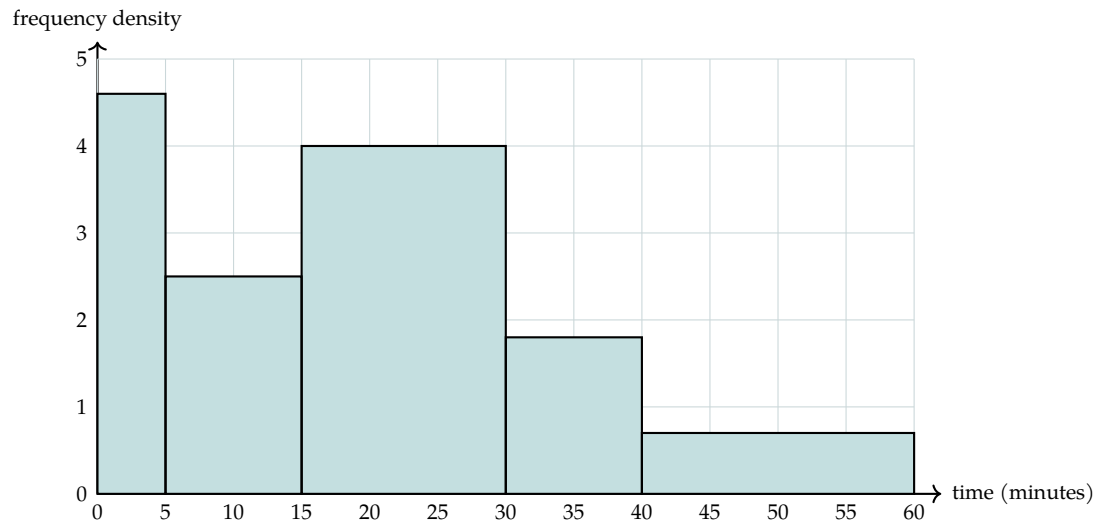
Frequencies 23, 25, 60, 18, 14; densities 4.6, 2.5, 4, 1.8, 0.7

Completed diagram 10

Complete a grouped table and histogram

MODULAR UNIT 1

4 marks

Completed solution diagram: Completed histogram

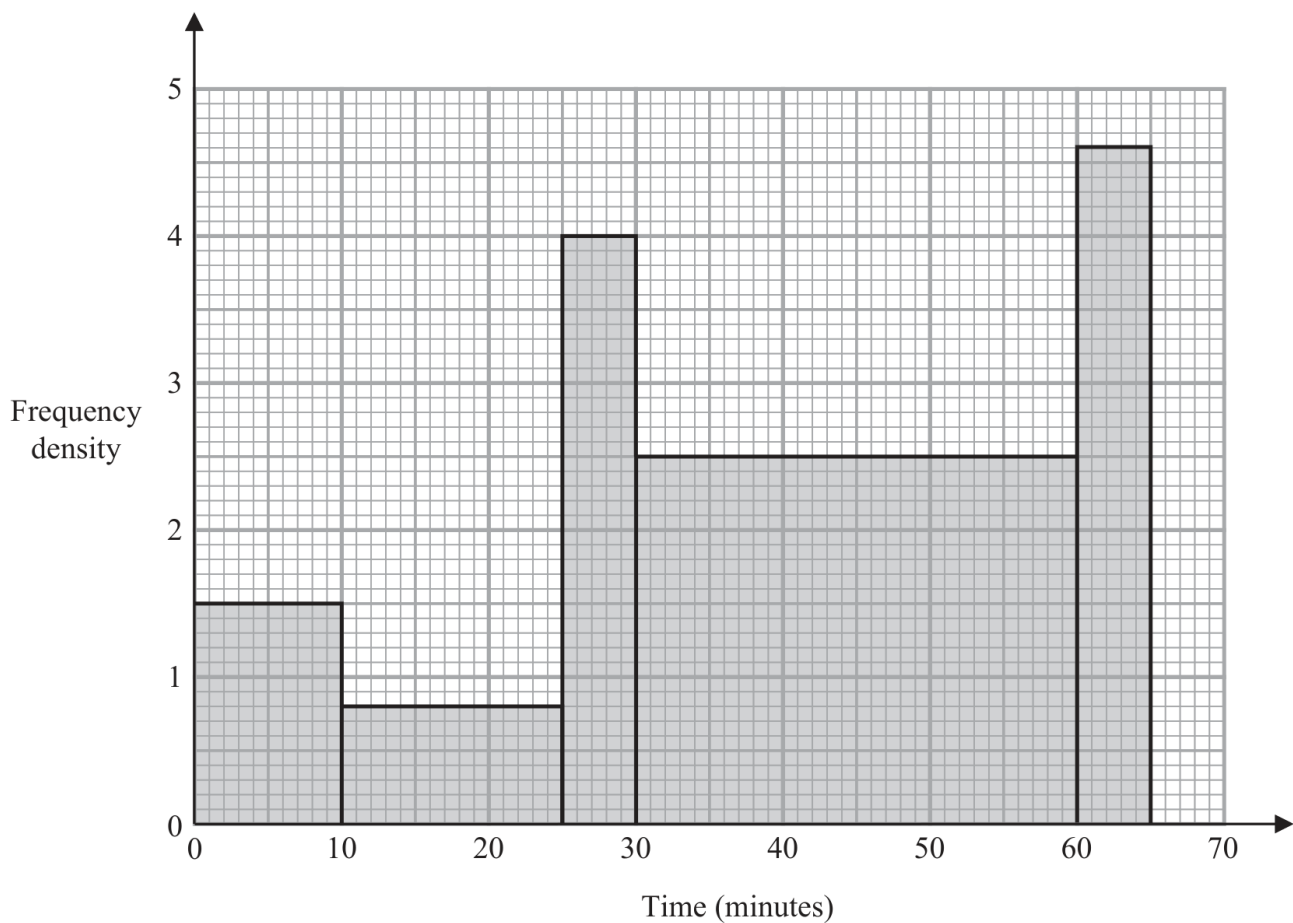
Original question 15

4MA1-1H Q18 M01

MODULAR UNIT 1

4 marks

- 18 The histogram shows information about the times some students took to complete a puzzle.



Work out an estimate for the fraction of these students who took between 20 minutes and 60 minutes to complete the puzzle.

Worked solution 15

Estimate a histogram fraction

MODULAR UNIT 1

4 marks

M01 – Estimate a histogram fraction**Find the interval frequencies**Use frequency density $\times$ class width: the total frequency is 145.**Find the target frequency**Between 20 and 60 minutes: $20 + 75 + 4 = 99$.**Form the fraction**

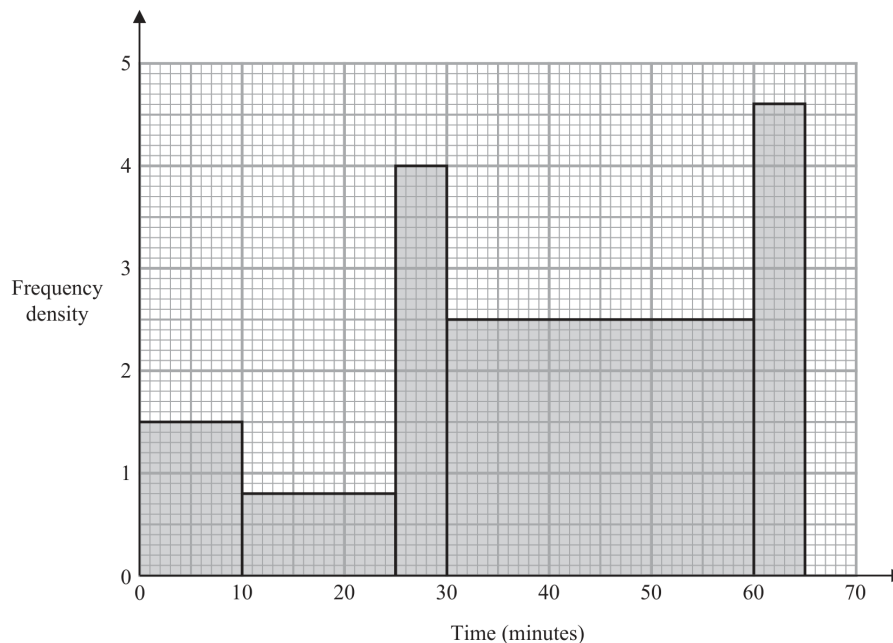
$$\text{fraction} = 99/145.$$

Final answer

$$\frac{99}{145}$$

Completed solution diagram: Histogram-area calculation on the exact source scale

- 18 The histogram shows information about the times some students took to complete a puzzle.



Work out an estimate for the fraction of these students who took between 20 minutes and 60 minutes to complete the puzzle.

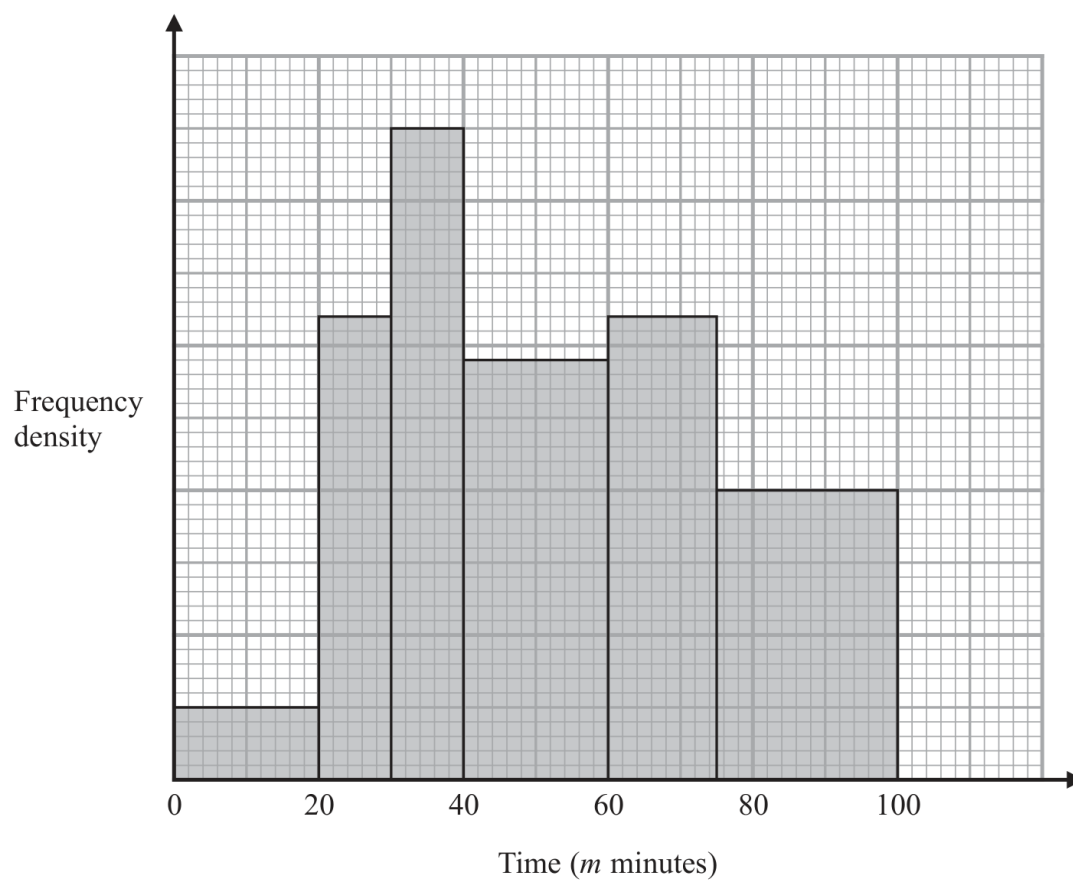
Target classes from 20 to 60 minutes: $20 + 75 + 4 = 99$; total frequency 145.

Original question 14

4MA1-1H Q21 Q21

ORIGINAL QUESTION**3 marks**

- 21 The histogram shows information about the total time, m minutes, taken by each child in a school to walk to school every day for one week.



There are no children for whom $m > 100$

There are 10 children for whom $m \leq 20$

Work out an estimate for the number of children for whom $50 < m \leq 80$

Worked solution 14

Chapter: Histograms

MODULAR UNIT 1

3 marks

Q21 – Chapter: Histograms

Family: Interpret histogram areas • Variant: Estimate a partial-class frequency

Method

For $0 < m \leq 20$, frequency density = $10/20 = 0.5$; use this bar to calibrate the vertical scale.

Why: Histogram area equals frequency, and density equals frequency divided by class width.

Method

From the calibrated grid, the densities over 50-60, 60-75, and 75-80 are approximately 2.9, 3.2, and 2.0.

Why: Split the requested interval wherever a histogram class boundary occurs.

Method

Estimated frequency = $10(2.9) + 15(3.2) + 5(2.0) = 29 + 48 + 10 = 87$ children.

Why: Add the areas of the three relevant histogram pieces.

Final answer

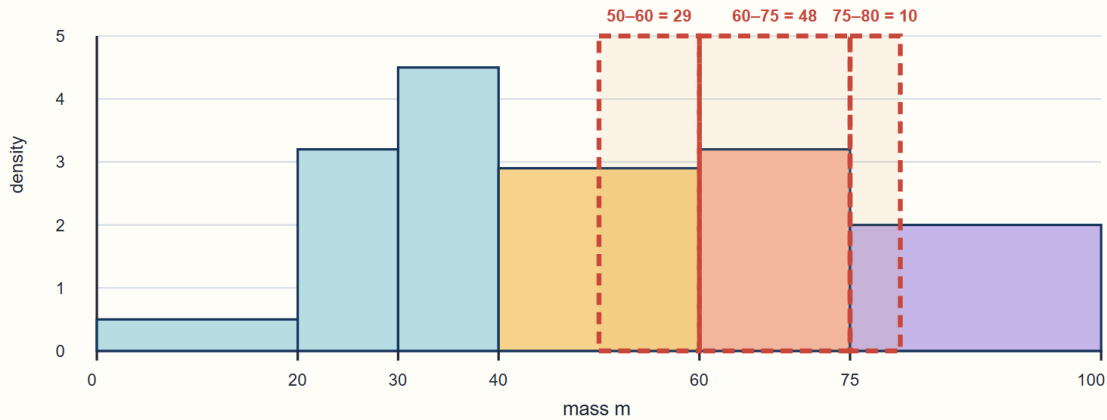
estimated children: 87

Completed diagram 14

Chapter: Histograms

MODULAR UNIT 1

3 marks

Q21(a) Annotated histogram readingDensity $\times$ class width gives frequency; split 50–80 at 60 and 75Target estimate: $10 \times 2.9 + 15 \times 3.2 + 5 \times 2.0 = 87$ Check: frequency = density $\times$ width

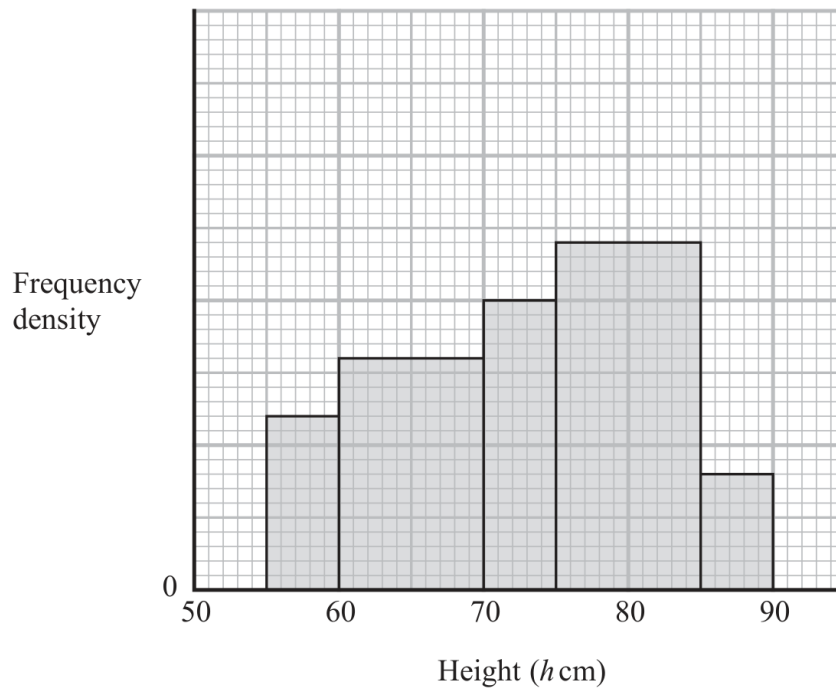
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Original question 26

4MA1-2H Q20

ORIGINAL QUESTION**4 marks**

20 The histogram gives information about the heights, h cm, of some tomato plants.



There are 12 tomato plants for which $75 < h \leq 85$

One of the tomato plants is selected at random.

Find an estimate for the probability that this tomato plant has a height greater than 82.5 cm

Worked solution 26

Chapter: Histograms

MODULAR UNIT 1

4 marks

Q20 – Chapter: Histograms

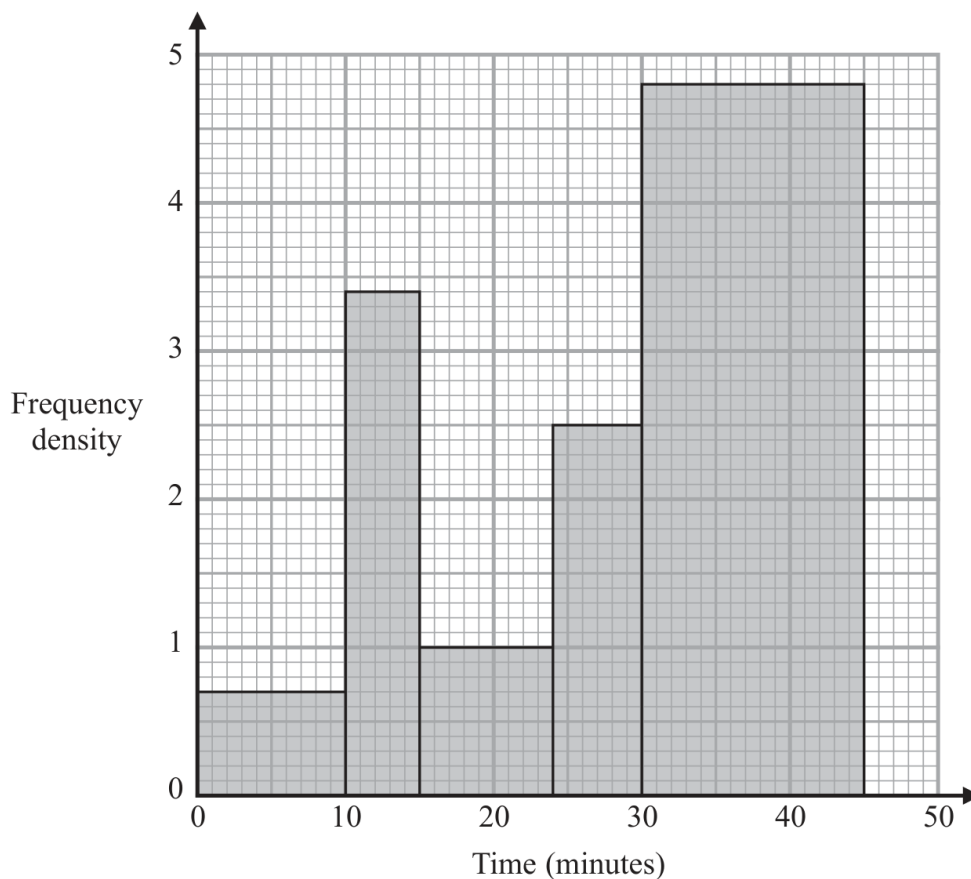
Family: Interpret histogram areas • Variant: Estimate a histogram proportion, fraction or percentage

Method

The frequency densities are 0.6, 0.8, 1.0, 1.2 and 0.4, giving frequencies 3, 8, 5, 12 and 2.

*Why: Frequency equals class width multiplied by density.***Method**Total frequency = $3 + 8 + 5 + 12 + 2 = 30$.*Why: Add all class frequencies.***Method**Above 82.5 cm: half of the 75–85 class contributes $2.5 \times 1.2 = 3$, plus the 85–90 class contributes 2.*Why: Use the histogram area proportion.***Method**Probability = $5/30 = 1/6$.*Why: Divide the favourable frequency by 30.***Final answer**probability: $1/6$

- 18 The histogram gives information about the times, in minutes, that some customers spent in a supermarket.



- (a) Work out an estimate for the proportion of these customers who spent between 17 minutes and 35 minutes in the supermarket.

(3)

Worked solution

January 2020 | Question 18 | 3 marks

Family: Interpret histogram areas • Variant: Estimate a histogram proportion, fraction or percentage

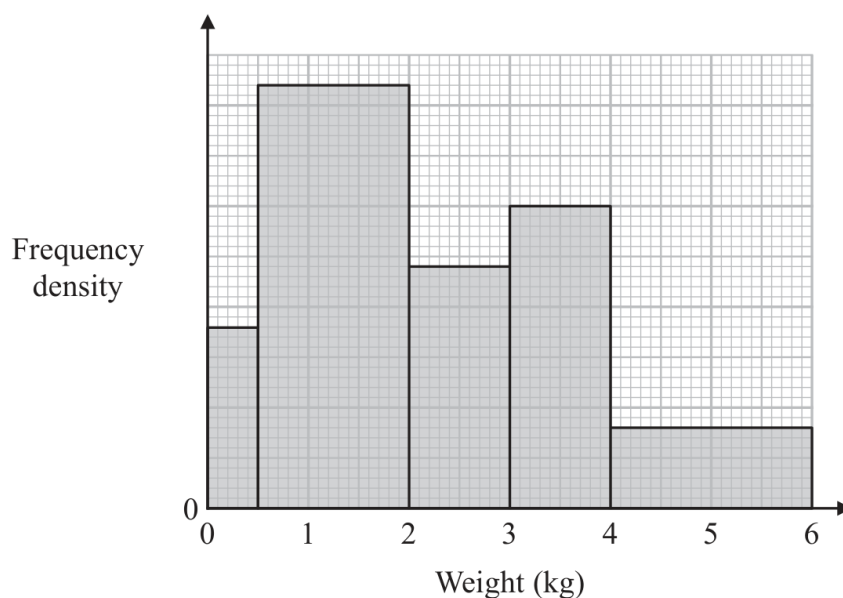
Method / working

1. Total frequency=120 from the histogram areas.
2. Frequency between 17 and 35=46.
3. Proportion= $46/120=23/60$.

Final answer: $23/60$

15 A postman records the weight of each parcel that he delivers.

The histogram shows information about the weights of all the parcels that the postman delivered last Monday. No parcels weighed more than 6 kg.



63 of the parcels that the postman delivered last Monday each had a weight between 0.5 kg and 2 kg.

(a) Work out the total number of parcels the postman delivered last Monday.

.....
(3)

The postman picks at random two of the records of the parcels he delivered last Monday.

(b) Work out an estimate for the probability that each parcel weighed more than 2.25 kg.

.....
(3)

Worked solution

January 2021 | Question 15 | 3 marks

Family: Use the frequency-density model • Variant: Calibrate an unnumbered frequency-density scale

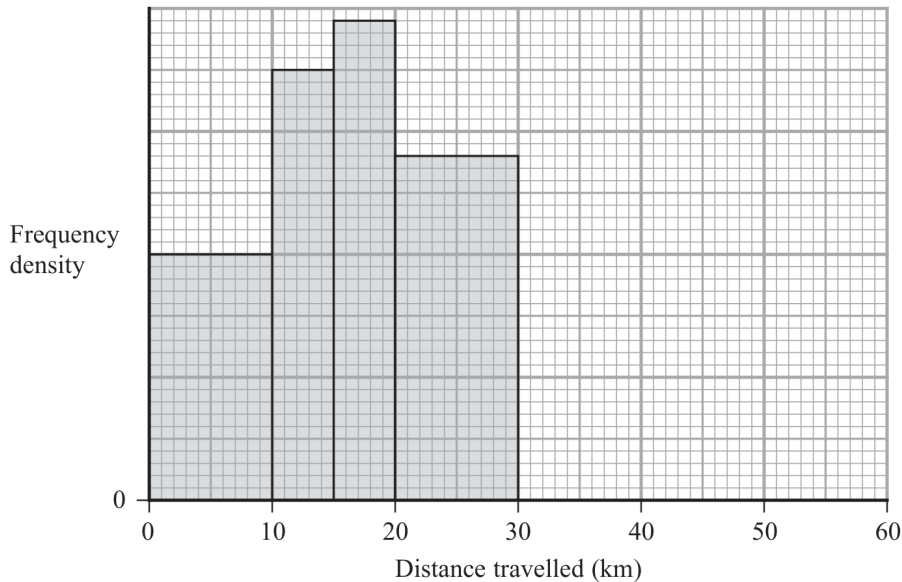
Method / working

1. Use the given 63 parcels to calibrate the histogram scale.
2. Sum the five bar areas on the frequency-density scale.
3. The total frequency is 142 parcels.

Final answer: 142 parcels

- 18 The table and histogram give information about the distance travelled, in order to get to work, by each person working in a large store.

Distance (d km)	Frequency
$0 \leq d < 10$	40
$10 \leq d < 15$	
$15 \leq d < 20$	
$20 \leq d < 30$	
$30 \leq d < 60$	30



Using the information in the table and in the histogram,

- (a) complete the table,

(2)

- (b) complete the histogram.

(1)

One of the people working in the store is chosen at random.

- (c) Work out an estimate for the probability that the distance travelled by this person, in order to get to work, was greater than 25 km.

(2)

Worked solution

June 2021 | Question 18 | 2 marks

Family: Construct and complete histograms • Variant: Complete a table from a partial histogram

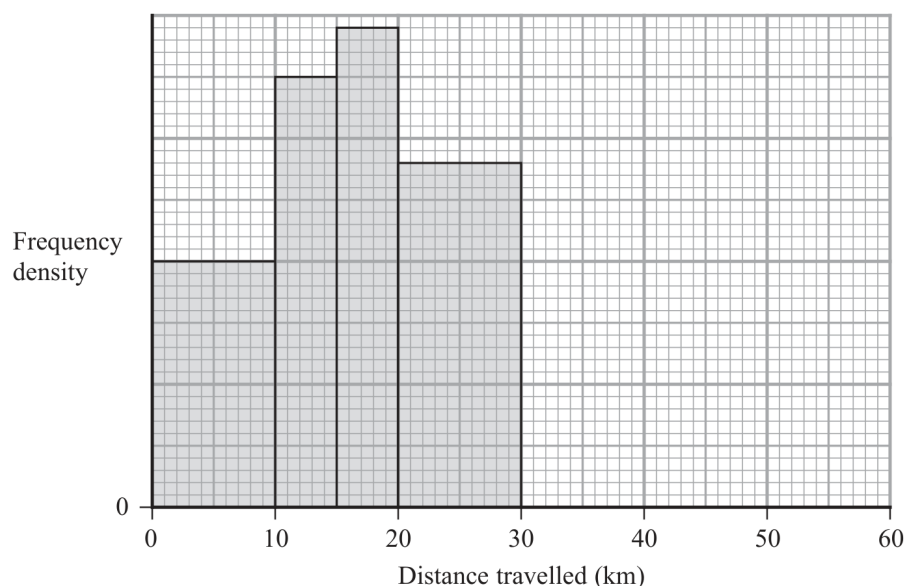
Method / working

1. Use frequency density = frequency / class width to complete the missing table values 35 and 39.
2. Read the remaining frequency from the histogram scale: 56.

Final answer: 35, 39, 56

- 18 The table and histogram give information about the distance travelled, in order to get to work, by each person working in a large store.

Distance (d km)	Frequency
$0 \leq d < 10$	40
$10 \leq d < 15$	
$15 \leq d < 20$	
$20 \leq d < 30$	
$30 \leq d < 60$	30



Using the information in the table and in the histogram,

- (a) complete the table,

(2)

- (b) complete the histogram.

(1)

One of the people working in the store is chosen at random.

- (c) Work out an estimate for the probability that the distance travelled by this person, in order to get to work, was greater than 25 km.

(2)

Worked solution

June 2021 | Question 18 | 1 marks

Family: Construct and complete histograms • Variant: Complete missing histogram bars from mixed information

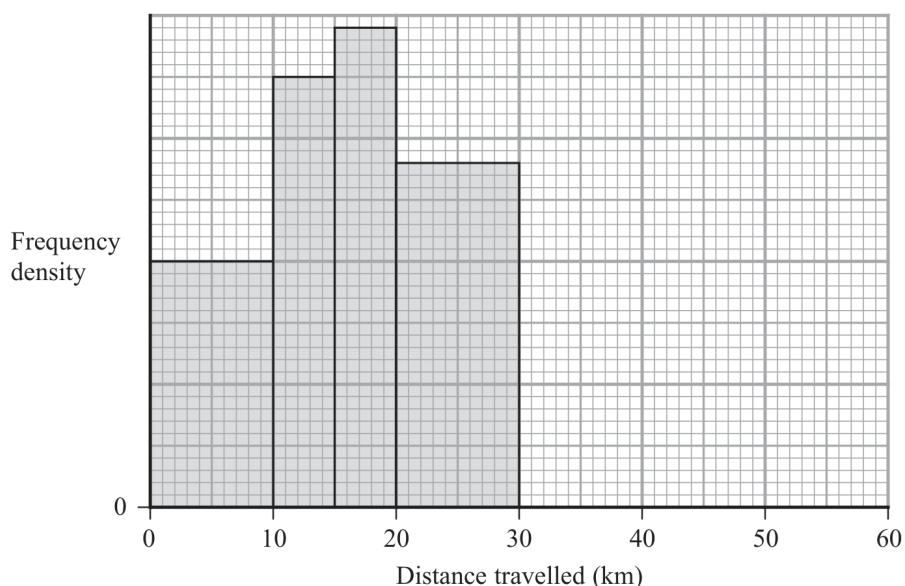
Method / working

1. Complete the 30-60 class with the bar height 1 on the frequency-density scale.

Final answer: Height 1 for $30 \leq d < 60$

- 18 The table and histogram give information about the distance travelled, in order to get to work, by each person working in a large store.

Distance (d km)	Frequency
$0 \leq d < 10$	40
$10 \leq d < 15$	
$15 \leq d < 20$	
$20 \leq d < 30$	
$30 \leq d < 60$	30



Using the information in the table and in the histogram,

- (a) complete the table,

(2)

- (b) complete the histogram.

(1)

One of the people working in the store is chosen at random.

- (c) Work out an estimate for the probability that the distance travelled by this person, in order to get to work, was greater than 25 km.

(2)

Worked solution

June 2021 | Question 18 | 2 marks

Family: Interpret histogram areas • Variant: Estimate a histogram proportion, fraction or percentage

Method / working

1. Estimated count above 25 km = $0.5 \times 56 + 30 = 58$ (the 30-60 class contributes 30; do not add the already-used table frequencies again).

2. Total count = 200, so probability = $58/200 = 0.29$.

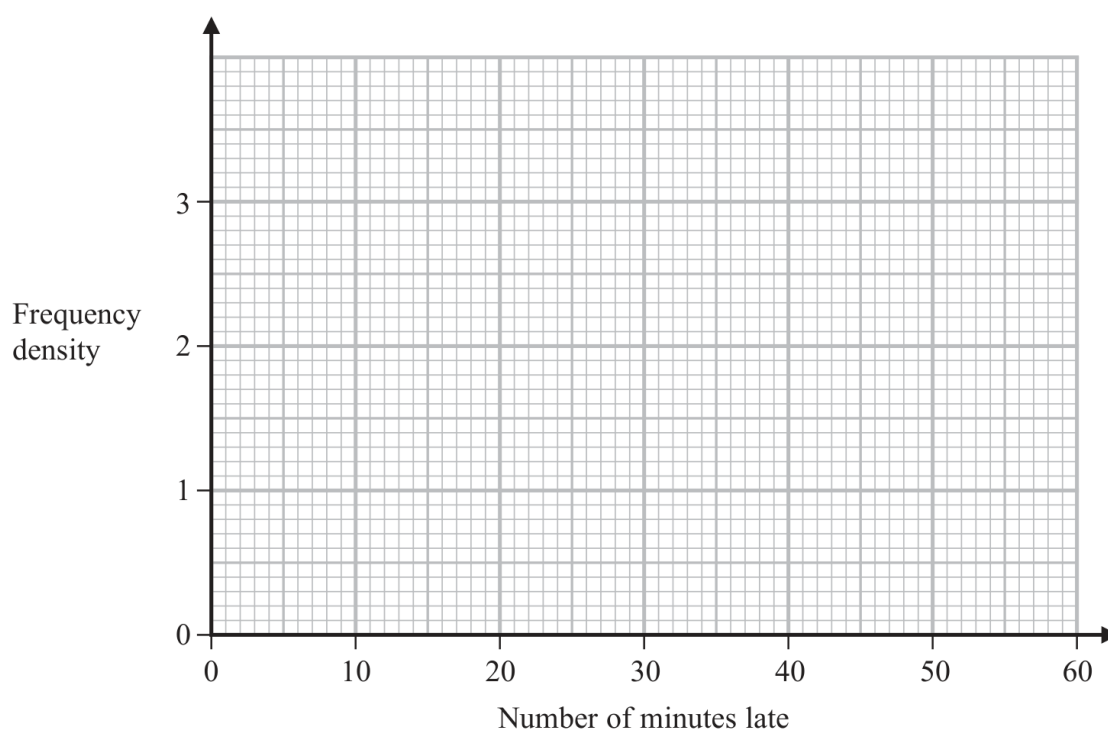
Final answer: 0.29

14 Manuel collected information about the flights that arrived late at an airport last month.

The table gives information about the number of minutes that these flights were late.

Minutes late (L minutes)	Frequency
$0 < L \leq 10$	8
$10 < L \leq 15$	13
$15 < L \leq 25$	19
$25 < L \leq 40$	24
$40 < L \leq 60$	6

(a) On the grid, draw a histogram for this information.



(3)

Manuel selected at random a flight that was late by 25 minutes or less from his results.

(b) Work out an estimate for the probability that this flight was late by 5 minutes or less.

(2)

Worked solution

November 2021 | Question 14 | 3 marks

Family: Construct and complete histograms | Variant: Draw a histogram from a grouped table

METHOD / WORKING

1. The histogram densities are

$$2.0 < L \leq 10 : \frac{8}{10} = 0.8$$

$$3.10 < L \leq 15 : \frac{13}{5} = 2.6$$

$$4.15 < L \leq 25 : \frac{19}{10} = 1.9$$

$$5.25 < L \leq 40 : \frac{24}{15} = 1.6$$

$$6.40 < L \leq 60 : \frac{6}{20} = 0.3$$

7. Given $L \leq 25$, the total frequency is

$$8. 8+13+19=40$$

9. For $L \leq 5$, estimate half of the first class:

$$10. 5(0.8)=4$$

$$11. P(L \leq 5 | L \leq 25) = \frac{4}{40} = \frac{1}{10}$$

12. Answer: histogram densities 0.8, 2.6, 1.9, 1.6, 0.3, and probability $\frac{1}{10}$.

FINAL ANSWER

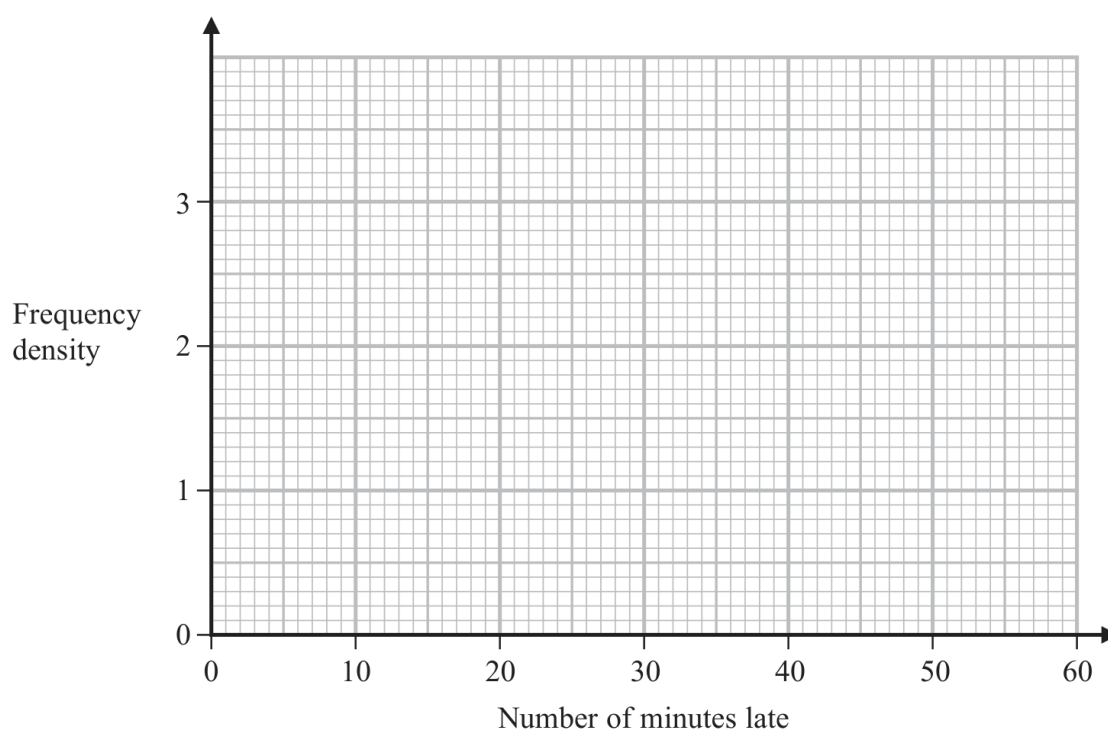
histogram densities 0.8, 2.6, 1.9, 1.6, 0.3, and probability $\frac{1}{10}$.

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$10 < L \leq 15$	13
$15 < L \leq 25$	19
$25 < L \leq 40$	24
$40 < L \leq 60$	6

(a) On the grid, draw a histogram for this information.



(3)

Manuel selected at random a flight that was late by 25 minutes or less from his results.

(b) Work out an estimate for the probability that this flight was late by 5 minutes or less.

(2)

Worked solution

November 2021 | Question 14 | 2 marks

Family: Interpret histogram areas | Variant: Estimate a histogram proportion, fraction or percentage

METHOD / WORKING

1. The histogram densities are

$$2.0 < L \leq 10 : \frac{8}{10} = 0.8$$

$$3.10 < L \leq 15 : \frac{13}{5} = 2.6$$

$$4.15 < L \leq 25 : \frac{19}{10} = 1.9$$

$$5.25 < L \leq 40 : \frac{24}{15} = 1.6$$

$$6.40 < L \leq 60 : \frac{6}{20} = 0.3$$

7. Given $L \leq 25$, the total frequency is

$$8. 8+13+19=40$$

9. For $L \leq 5$, estimate half of the first class:

$$10. 5(0.8)=4$$

$$11. P(L \leq 5 | L \leq 25) = \frac{4}{40} = \frac{1}{10}$$

12. Answer: histogram densities 0.8, 2.6, 1.9, 1.6, 0.3, and probability $\frac{1}{10}$.

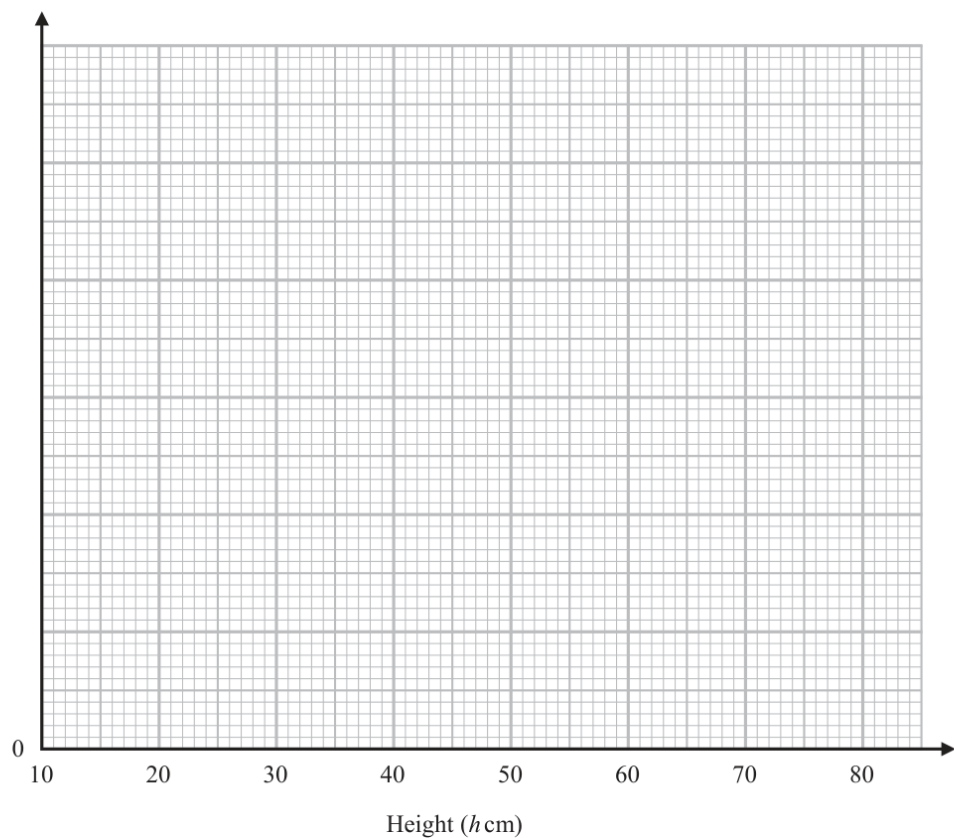
FINAL ANSWER

histogram densities 0.8, 2.6, 1.9, 1.6, 0.3, and probability $\frac{1}{10}$.

18 The table gives information about the heights, in centimetres, of some plants.

Height (h cm)	Frequency
$10 < h \leq 20$	35
$20 < h \leq 35$	45
$35 < h \leq 50$	75
$50 < h \leq 70$	40
$70 < h \leq 80$	8

(a) On the grid, draw a histogram for this information.



(3)



(b) Work out an estimate for the number of these plants with a height greater than 40 cm.

.....
(2)

(Total for Question 18 is 5 marks)

Answer reference | May-Nov 2020 | Question 18

Family: Interpret histogram areas • Variant: Estimate a histogram proportion, fraction or percentage

18	(a)	$35 \div 10 (=3.5)$, $45 \div 15 (=3)$, $75 \div 15 (=5)$, $40 \div 20 (=2)$, $(8 \div 10) = 0.8$		3	M1	for any two correct fd or two correct bars drawn of different widths
		$35 \div 10 (=3.5)$ and $45 \div 15 (=3)$ and $75 \div 15$ $(=5)$ and $40 \div 20 (=2)$ and $(8 \div 10) = 0.8$			M1	for all correct fd or at least 3 correct bars drawn
					A1	for a fully correct histogram with 'frequency density' (or fd) and scale on the axis labelled or appropriate key (SC: B2 for all five bars drawn of correct width with heights in the correct ratio) (SC: B1 for three bars drawn of correct width with heights in the correct ratio)
	(b)	$10 \times 5 + 40 + 8$ or $\frac{2}{3} \times 75 + 40 + 8$		2	M1	ft from their histogram in (a) for a correct method
			98		A1	
						Total 5 marks

Worked solution

May-Nov 2020 | Question 18 | 5 marks

Family: Interpret histogram areas • Variant: Estimate a histogram proportion, fraction or percentage

Solution

The histogram densities are

$$(35)/(10)=3.5, \text{quad}$$

$$(45)/(15)=3, \text{quad}$$

$$(75)/(15)=5, \text{quad}$$

$$(40)/(20)=2, \text{quad}$$

$$(8)/(10)=0.8$$

For plants taller than 40 cm:

From 40<h≤ 50:

$$10(5)=50$$

From 50<h≤ 70:

$$40$$

From 70<h≤ 80:

$$8$$

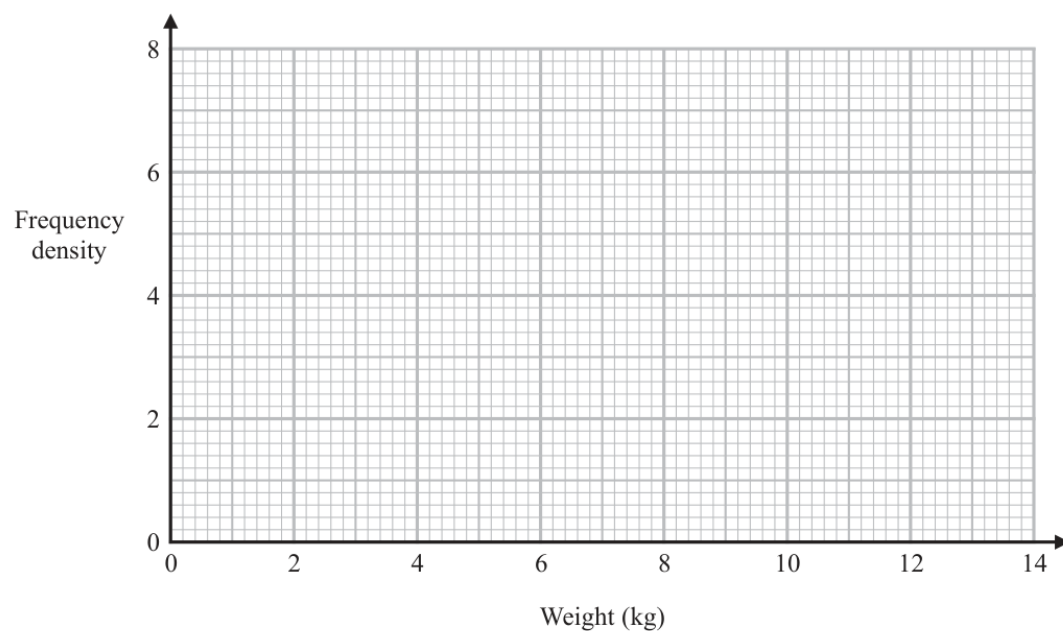
$$50+40+8=98$$

Answer: draw bars with densities 3.5, 3, 5, 2, 0.8, and 98 plants.

- 18 The table gives information about the weights, in kg, of the parcels that Pedro delivers on Monday.

Weight (w kg)	Frequency
$0 < w \leq 2$	12
$2 < w \leq 3$	7
$3 < w \leq 6$	15
$6 < w \leq 9$	12
$9 < w \leq 14$	9

- (a) On the grid, draw a histogram for this information.



(3)

One of the parcels that Pedro delivered on Monday is chosen at random.

- (b) Using the information in the table, find an estimate for the probability that this parcel weighs more than 7 kg.

(2)

(Total for Question 18 is 5 marks)

Answer reference | January 2023 | Question 18

Family: Interpret histogram areas • Variant: Estimate a histogram proportion, fraction or percentage

USE OVERLAY PROVIDED				
18	(a)	FD are: 6, 7, 5, 4, 1.8	3	M1 For at least two frequency densities correct or at least two correct bars
				M1 For at least 4 correct frequency densities or 4 correct bars
		<i>A fully correct histogram gains full marks</i>	Correct histogram	A1 Fully correct histogram SCB2 for all five bars of correct width with heights in the correct ratio (eg drawn at 0.6, 0.7, 0.5, 0.4, 0.18) SCB1 for three bars of correct width with heights in the correct ratio
	(b)	$(9 + \frac{2}{3} \times 12) (= 17)$ oe eg $9 + 8 (= 17)$ or $55 - (12 + 7 + 15 + \frac{1}{3} \times 12)$	2	M1 may be seen as numerator of fraction (ft their graph dep on M1 in (a))
		<i>Correct answer scores full marks (unless from obvious incorrect working)</i>	$\frac{17}{55}$	A1cao Or 0.30909...or 30.909...% (to at least 2 sf) SCB1 for $\frac{38}{55}$ (0.6909...)
Total 5 marks				

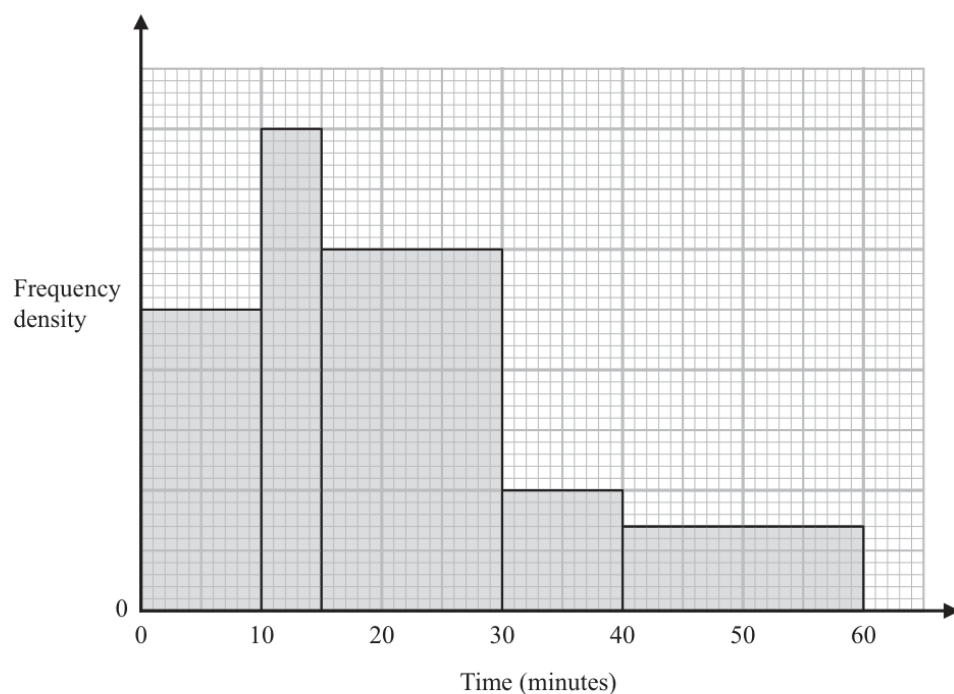
Worked solution

January 2023 | Question 18 | 5 marks

Family: Interpret histogram areas • Variant: Estimate a histogram proportion, fraction or percentage

Official final mark-scheme pages are embedded immediately after this question.

- 21 The histogram shows information about the times, in minutes, that some trains arrived late at a station one day.



20 of these trains arrived between 10 minutes late and 15 minutes late.

No trains arrived more than 60 minutes late.

Work out an estimate for the number of these trains that arrived at least 25 minutes late.

(Total for Question 21 is 3 marks)

Worked solution

November 2025 | Question 21 | 3 marks

Family: Interpret histogram areas • Variant: Estimate a histogram proportion, fraction or percentage

Method

For a histogram,

frequency = class width $\times$ frequency density

For $10 < t \leq 15$, the frequency is 20 and the class width is 5, so

frequency density = $(20)/(5) = 4$

Using this scale from the histogram:

$15 < t \leq 30$ quad has frequency density 3

$30 < t \leq 40$ quad has frequency density 1

$40 < t \leq 60$ quad has frequency density 0.7

For trains at least 25 minutes late:

$(5)(3) + (10)(1) + (20)(0.7)$

$= 15 + 10 + 14$

$= 39$

Answer: 39 trains